

Ali Nikkhah

Tehran, Iran | +98 991 296 3951 | alinkkh9@gmail.com | alinikkhah2001.github.io | linkedin.com/in/alinikkhah2001

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, JAX, TensorFlow, Hugging Face, Time-Series, ONNX Runtime
Agentic AI & Advanced RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, Graph RAG, KAG
Vision & Audio Systems: OpenCV, BLIP, Vision Transformers (ViT), T5, Whisper, wav2vec2.0, Librosa, ESPnet
Data Engineering & Lakes: Apache Spark, Trino, ClickHouse, Airflow, PeerDB, Apache Kafka, RabbitMQ, Parquet
MLOps & Infrastructure: Ray, Kubeflow, Triton Inference Server, TensorFlow Serving, TensorRT-LLM, TF Lite
DevOps & Automation: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, GitLab CI
Monitoring, Governance & BI: Prometheus, Grafana, ELK Stack, Metabase, Weights & Biases, MLflow, DVC

PROFESSIONAL EXPERIENCE

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer

Sep 2025 - Mar 2026

- Designed, provisioned, and optimized the central enterprise data lake and data warehousing infrastructure, handling high-volume transaction routing and user interaction data streams generated by BNPL and stock platforms.
- Orchestrated complex, dependency-mapped workflows using Apache Airflow and deployed high-speed Change Data Capture (CDC) replication clusters via PeerDB to continuously mirror transactional databases into OLAP engines.
- Built automated, real-time statistical inference engines to execute parallel A/B hypothesis tests on application features, monitoring user activity telemetry and logging performance shifts under volatile market events.
- Refactored legacy data manipulation scripts into modular, object-oriented PySpark blocks, drastically improving memory management and reducing algorithmic processing complexities over data lake targets.
- Architected data transformation DAGs that automate cold-data archiving procedures, compressing historical relational records into optimized, immutable Parquet structures on object storage layers.
- Designed enterprise monitoring configurations combining Prometheus and Grafana alerts to trace pipeline latency spikes, pipeline failures, and storage volume degradation patterns.

Digikala

Tehran, Iran

Senior AI Engineer

Jan 2025 - Aug 2025

- Led the optimization of large-scale vector indexing layers, configuring Elasticsearch, OpenSearch, and FAISS infrastructures to execute fast hybrid searches across millions of unstructured corporate documents.
- Engineered memory-efficient document parsing, incremental asset indexing, and automated data synchronization workflows to ensure real-time consistency between core databases and downstream vector spaces.
- Designed distributed feature extraction jobs using Apache Spark to pre-compute, partition, and serialize massive text datasets prior to running batch embedding generation workflows.
- Built scalable ingestion pipelines capable of extracting, filtering, and tokenizing raw historical transaction data streams to pre-populate document indices.

HomaCloud

Tehran, Iran

MLOps Engineer

May 2021 - Feb 2022

- Managed distributed data pipelines and compute resources over Kubernetes using Ray and MLflow to automate multi-node model execution and feature extraction routines.
- Built automated data workflows for multi-node model training environments, ensuring clean horizontal data partitioning and data lineage tracking across ephemeral storage blocks.

SELECTED PROJECTS & RESEARCH

AI-Driven BNPL Risk Assessment & Credit Engine

Turquoise Digital

Senior Data Engineer

2025

- Developed and productionized predictive machine learning models to analyze user historical debt repayment structures, calculate credit limits, and dynamically determine automated account funding or containment blocks.
- Implemented structured data serialization formats (Parquet, ORC) within PySpark processing blocks to optimize data compression and query execution speeds across the central data lake.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering - Digital Systems (GPA: 3.65, Major: 3.91)

Sept 2020 - Jul 2024